



Deep learning based flood segmentation: evaluating EfficientNet and ResNet for UNet, SegNet and DeepLabV3+

Şeyda Karıcı¹ · Elif Özceylan Aslan² · Kazım Yıldız¹ · Önder Demir¹

Received: 25 February 2025 / Accepted: 7 January 2026 / Published online: 3 March 2026
© The Author(s) 2026

Abstract

Rapid and precise flood extent mapping is vital for effective disaster response, as floods cause infrastructure damage, crop losses, casualties, and major economic disruptions. Conventional hydrological models and remote sensing approaches often suffer from high computational demands, coarse spatial resolution, and limited real-time adaptability. This study offers a systematic comparison of three state-of-the-art deep learning segmentation architectures (UNet, SegNet, and DeepLabV3+), each paired with either EfficientNet or ResNet backbones. Our goal is to identify the most efficient and accurate model for flood-affected region delineation. In experiments, UNet with EfficientNet backbone outperformed alternatives, achieving an accuracy of 0.9705, a precision of 0.9609, a recall of 0.9548, a F1-score of 0.9575 and an IoU of 0.9271. These results highlight EfficientNet's strong feature extraction and lightweight design, making it ideal for near real-time operational deployment. Overall, our findings offer a practical, scalable solution for real-world disaster management, enabling faster damage assessment, optimized resource allocation, and improved flood response.

Keywords Flood segmentation · UNet · ResNet · SegNet · EfficientNet · DeepLabV3+

✉ Elif Özceylan Aslan
elifozceylanaskan@topkapi.edu.tr

Şeyda Karıcı
seyda.karci@marmara.edu.tr

Kazım Yıldız
kazim.yildiz@marmara.edu.tr

Önder Demir
odemir@marmara.edu.tr

¹ Computer Engineering Department, Technology Faculty, Marmara University, Istanbul, Türkiye

² Faculty of Engineering and Natural Sciences, Computer Engineering Department, Istanbul Topkapı University, Istanbul, Türkiye

1 Introduction

Floods are among the most frequent natural disasters in the world, comprising between 40% and 50% of all disasters. They also inflict the greatest number of fatalities and financial damages globally. Analysis between 2000 and 2009 shows that floods account for 38.7% of all natural disasters, are responsible for 6.2% of disaster-related deaths, and cover 43% of the total affected population. These data are presented based on the EM-DAT database and reveal the magnitude of the health and economic impacts of floods on a global scale (Du et al. 2010). Rapidly increasing urbanization also increases the population density and infrastructure exposure in flood-prone areas, significantly increasing the economic and structural losses caused by floods. Hemmati et al. emphasized in their study how socio-economic development and consequently urbanization change the fundamental elements of risk (vulnerability, exposure, and hazard) in flood-prone areas and suggested that this trend will continue to increase in the future (Hemmati et al. 2020). In addition to the increasing population, various factors such as excessive rainfall, storm waves, rapid snow melting, and sea level rise trigger flood events. Kumar et al. examined different flood modeling methods in their studies and emphasized that hydrological and hydraulic models simulate rainfall-run-off processes, numerical modeling techniques (such as HEC-RAS and MIKE FLOOD) calculate water flow, Geographic Information Systems (GIS) and remote sensing-based models map flood areas using satellite images. In addition, Artificial Intelligence (AI) and Machine Learning (ML) methods (such as CNN, LSTM, and ANN) provide more accurate flood predictions with big data analysis, and MCDA, or multi-criteria decision analysis methods, evaluate flood management strategies, determine the effectiveness of structural (levee, dam) and non-structural (early warning systems, land management) measures. The study points out that the combination of AI and big data can increase the accuracy of flood forecasts and that interdisciplinary approaches will help create more thorough flood management plans by tackling current issues like data scarcity, model uncertainties, and computational complexity (Kumar et al. 2023). Advancements in Deep Learning (DL) have further enhanced flood modeling by enabling automated, high-accuracy water segmentation and real-time flood monitoring, which are essential for effective disaster response. Traditional flood detection technologies frequently rely on manual observations and semi-automated processes, limiting their efficiency and scalability. In addition, mapping flood-prone areas is challenging due to complex terrain, heterogeneous land cover, and varying environmental conditions. Notably, maps based on a single image often lead to inaccurate predictions, as non-flood pixels may be misclassified as flooded pixels. Such water-like appearances typically include smooth surfaces such as sand and asphalt, dry and sparse vegetation, bare rock surfaces, and high mountain ranges (Bauer-Marschallinger et al. 2022). To address these limitations, recent studies have focused on leveraging DL architectures such as UNet, ResNet, and DeepLabV3, which can process large-scale image datasets obtained from satellites, drones, and social media sources to segment flooded areas accurately. UNet improves floodwater discrimination by preserving fine details in low-contrast regions through skip connections. SegNet, on the other hand, reduces information loss in its encoder-decoder structure by utilizing max-pooling indices, which is particularly beneficial in scenarios involving surface-water ambiguities, such as on smooth terrains. DeepLabV3+ extracts multi-scale contextual information through Atrous Spatial Pyramid Pooling (ASPP), enabling the model to better handle various land cover types and environmental complexities. ResNet provides a deeper

network architecture, which is advantageous in distinguishing regions with subtle differences (e.g., between flooded and dry grass areas). It also offers higher levels of abstraction, which aids in extracting details in radar shadow regions. EfficientNet optimizes network width, depth, and resolution simultaneously through compound scaling without increasing the number of parameters. This makes it effective for single-image-based segmentation tasks by achieving better performance with less data. The use of UNet, SegNet, and DeepLabV3+ architectures in combination with ResNet and EfficientNet backbones is expected to improve boundary precision in segmentation, reduce information loss, and enhance generalization through optimized parameter usage. This study aims to identify the most effective segmentation architecture-backbone combination for flood mapping. Our contributions include an examination of model performance in terms of accuracy, precision, recall, F1-score and IoU; a comparison of EfficientNet and ResNet as backbone networks; and a thorough evaluation of deep learning-based segmentation models for flood segmentation. Experimental results indicate that UNet with EfficientNet outperforms other configurations, achieving the highest segmentation accuracy. These findings demonstrate the potential of DL models to enhance flood mapping and disaster response strategies.

2 Related work

Techniques from AI and ML have drawn a lot of interest lately for jobs involving segmentation and flood prediction. Traditional hydrological and hydraulic models, while accurate, often suffer from computational inefficiencies, limiting their applicability in real-time flood monitoring. Similarly, approaches based on remote sensing and Geographic Information Systems (GIS), despite their spatial analysis capabilities, frequently require manual intervention, reducing their scalability and efficiency. To overcome these challenges, DL approaches have emerged as powerful alternatives, leveraging large-scale image data to improve the accuracy and automation of flood mapping. Convolutional Neural Networks (CNN), Multilayer Perceptrons (MLP), and Recurrent Neural Networks (RNN) are a few of the DL architectures that have been investigated for flood-related applications. Studies indicate that CNN-based models achieve high accuracy in flood segmentation tasks, demonstrating their superior capability in extracting spatial features from satellite and aerial imagery. Key applications include flood susceptibility analysis, flood risk mapping, and real-time flood risk assessment, where DL models have shown promising results in improving disaster management and mitigation efforts (Bentivoglio et al. 2022). AI-driven flood prediction systems use remote sensing, meteorological data, and geospatial analysis to improve risk assessment and decision-making processes (Caleb and Adeola 2024).

Several ML approaches have been employed for flood prediction and segmentation. Flood susceptibility mapping has made extensive use of Support Vector Machines (SVM) and Random Forest (RF) models, which analyze past flood episodes and environmental characteristics (Ganjirad and Delavar 2023). Artificial Neural Networks (ANN) have also been applied to predict flood occurrence based on meteorological and hydrological parameters, offering improved predictive accuracy compared to traditional statistical methods (Chebii et al. 2022). Deep learning-based segmentation models have demonstrated remarkable performance in flood detection. UNet, a popular CNN-based architecture, has been extensively used for semantic segmentation of water bodies in satellite and drone imag-

ery (Pech-May et al. 2023). Similarly, DeepLabV3 has been applied for UAV-based flood segmentation, using atrous convolutions to improve feature extraction and improve segmentation accuracy (Simantiris and Panagiotakis 2024). SegNet, another encoder-decoder architecture, has been explored for flood segmentation tasks, particularly in cases where computational efficiency is a priority (Bahrami and Arbabkhah 2024). Recent studies have investigated the use of EfficientNet and ResNet as backbone networks for feature extraction in segmentation models (Hernández et al. 2022).

Utilizing the publicly available Sentinel-1 dataset from NASA and IEEE GRSS, Ghosh et al. present a DL method for flood detection utilizing the UNet and Feature Pyramid Network (FPN) architectures with an EfficientNet-B7 backbone. They assessed the models' ability to generalize to other geographic regions, such as Bangladesh, Florence, North Alabama, and Nebraska. The results indicate that models trained on specific flood events can generalize well to new regions, with FPN achieving the highest segmentation accuracy. These findings highlight the potential of deep learning-based flood mapping approaches for improving large-scale disaster management systems (Ghosh et al. 2024).

Using publically available datasets, Chamatidis et al. describe a flood detection system that combines Image Transformers (ViTs) and transfer learning for satellite image processing. Their findings show that the proposed approach achieves 84.84% accuracy on Sentinel-1 SAR images and 83.14% accuracy on Sentinel-2 MSI images, demonstrating the robustness of the model across different image modalities. Furthermore, a comparative analysis against CNN-based architectures, including EfficientNet and ResNet, reveals that the ViT model performs up to 15% better on SAR images and 9% better on MSI images than typical CNN models. These results demonstrate that while CNN-based architectures provide strong baseline performance for flood mapping, Image Transformers offer superior feature extraction capabilities, especially in capturing multi-scale spatial dependencies required for flood risk assessment and management (Chamatidis et al. 2024).

Patil et al. present a method to generate flood maps and assess flood damage using synthetic aperture radar (SAR) images from the Sentinel-1 satellite. They propose an ensemble model combining ResNet34, InceptionV3, and VGG16, which achieves an average Intersection Over Union (IoU) score of 75.84% and an F1-score of 0.76. The model is validated on a case study in Kolhapur, India, and demonstrates its effectiveness in analyzing flood impact across a variety of land use types, including urban, agricultural, and forested areas. These findings highlight the potential of deep learning-based ensemble models to improve flood risk assessment and disaster response (Patil et al. 2024).

Bahrami and Arbabkhah evaluate the performance of DL (SegNet, UNet, and FCN32) for automatic floodplain segmentation using a dataset of 290 images. Their findings show that SegNet achieves the highest segmentation accuracy with 88% precision and a Dice Coefficient of 0.84 in identifying water zones in flood-affected areas. This study highlights the effectiveness of DL in improving flood extent mapping and highlights the potential of segmentation models to enhance flood risk assessment and management (Bahrami and Arbabkhah 2024).

Nouwakpo et al. (2024) develop a deep learning tool based on a UNet model to map irrigation methods in agricultural areas using Landsat-derived images. They report an overall accuracy of 0.78, with precision values of 0.73 for flood, 0.82 for sprinkler, and 0.80 for other irrigation methods. This model provides a valuable tool for water resource manage-

ment, especially in regions where natural precipitation is low and irrigation methods vary (Nouwakpo et al. 2024).

Using a variety of ML algorithms on remote sensing imagery within the Google Earth Engine platform, Sellami and Rhinane present a novel method for mapping flash flood (FF) exposure in Tetouan, Morocco. They find that the SVM model performs best for LULC mapping, while the RF algorithm achieves 96% accuracy in FF susceptibility assessment. The final FF exposure map reveals that 2465 hectares and 198,913 inhabitants are at risk, demonstrating the model's potential for informing flood mitigation strategies in similar regions worldwide (Sellami and Rhinane 2024).

Munawar et al. present a comprehensive framework for categorizing recent advancements in flood management systems, systematically addressing methodologies, phases, and technologies employed in flood risk assessment and mitigation. They identify significant research gaps, including the limited integration of hybrid models that combine image processing and ML techniques, as well as the underutilization of ML applications in post-disaster response and recovery. The findings suggest that a more interdisciplinary approach, integrating disaster management principles with advanced computational methodologies, could improve flood management plans' efficacy and efficiency at every stage (Munawar et al. 2021).

Apene et al. apply the K-nearest neighbor algorithm to classify flood-prone areas within Nigeria using meteorological data on rainfall and temperature from 2014 to 2017. Their results indicate that specific regions are at higher flood risk based on historical weather patterns. This study demonstrates how data-driven approaches can help with proactive flood risk assessment and readiness (Apene et al. 2023).

Yolina and Rojali. enhance the Geomorphic Flood Area plugin by incorporating land cover data to improve flood-prone area classification. They calculate a runoff coefficient based on land cover and integrate it into the Geomorphic Flood Index, resulting in an improved F1-score for the modified plugin. This research demonstrates that combining flow accumulation with runoff coefficients increases the accuracy of flood-prone area classifications (Yolina and Rojali 2023).

Using Sentinel-2 satellite images captured both before and after the flash flood, Elstohy et al. concentrate on identifying and evaluating flood-affected areas between the coastal highways of Ras Ghareb and Zafaarana. To identify flooded pixels, they created an Image Difference Feature (IDF) model and used the Normalized Difference Water Index (NDWI) to categorize water bodies. The results demonstrated that their IDF model achieved a high accuracy of 98.9%, highlighting its effectiveness as a flood image processing technique and its potential in aiding decision-makers in crisis management (Elstohy and Ali 2023).

Kim et al. apply DL models to predict floodwater levels in the CarayCaray Basin on Bili-ran Island, Philippines, addressing limitations in the region's flood forecasting and warning systems. Their evaluation of predictive performance reveal that the deep neural network (DNN) model achieved the highest accuracy, with a normalized root mean square error (NRMSE) of 9.48. For flood risk classification, they implement extreme gradient boosting (XGBoost), RF, and decision tree models (DT), with XGBoost achieving the highest F1-score of 0.92. These findings highlight the potential of AI-driven methodologies in enhancing flood risk assessment and management strategies (Kim et al. 2023).

Sudiana et al. propose a novel approach for urban flood classification using synthetic aperture radar (SAR) data and a three-dimensional convolutional neural network (3DCNN)

that addresses the limitations of cloud cover during extreme rainfall events. Their study achieves a maximum flood area classification accuracy of 71.4% using multi-temporal Sentinel-1 SAR imagery over Jakarta, Indonesia. These findings highlight the potential of the 3DCNN model in improving early flood detection and urban flood risk assessment, providing a valuable tool for disaster mitigation and future applications (Sudiana et al. 2023).

Kalidhas and Sivakumar process pre and post flood Landsat satellite data using image processing techniques to assess the impact of floods on land use and land cover in Chengalpattu taluk, Tamil Nadu. They classify the area into five categories—water, urban, forest, barren land, and agriculture—using supervised classification techniques on a GIS platform, achieving overall classification accuracies of 94.25% in 2015 and 90.8% in 2016. Their results demonstrate that satellite data is effective in analyzing changes in land use and land cover due to floods through image classification techniques (Kalidhas and Sivakumar 2022).

Foroughnia et al. address the challenges of accurately delineating flood areas by developing a stepwise classification approach that combines unsupervised and supervised methods using SAR and multi-spectral (MS) data. In order to address heterogeneous surfaces, they combined supervised RF classification with a novel technique for unsupervised classification termed CThS (Combination of Thresholding and Segmentation). The results show that the CThS method achieves comparable precision and accuracy to RF, proving effective for rapid flood mapping and making it well-suited for operational use (Foroughnia et al. 2022).

Puttinaovarat et al. develop an application using DL for flood and land use classification, incorporating digital images, remote sensing, and crowdsourced data to improve flood impact assessment. They achieve a flood classification accuracy of 97.50% (Kappa = 0.95) and a land use classification accuracy of 93.71% (Kappa = 0.91), demonstrating the application's effectiveness in real-time flood damage visualization. Their approach, combining satellite data with user-reported damage, offers a novel and efficient method for assessing and visualizing flood impacts on a dashboard in real-time (Puttinaovarat et al. 2022).

Melgar-García et al. (2023) evaluate the performance of three CNN models—UNet, WUNet, and UNet++—for predicting pluvial flood-prone areas in a tropical region in northern Vietnam. They train these models using a geospatial dataset incorporating 12 flood indicators and past flood event data. The best-performing model achieves a high accuracy of 96.01%. The study's flood susceptibility maps give local governments important information that helps them make better decisions and implement efficient flood risk management plans (Melgar-García et al. 2023).

Dwiasnati et al. develop a flood-prone area model for Bandung Prefecture using the SVM algorithm, optimized with Particle Swarm Optimization (PSO) to enhance accuracy. They find that the PSO-based SVM model achieves an accuracy of 97.62% and an AUC of 1.000, significantly outperforming the standard SVM model, which has an accuracy of 85.71% and an AUC of 0.841. These results demonstrate the effectiveness of the optimized model in accurately classifying flood-prone and non-flood-prone areas, providing valuable information for the community (Dwiasnati and Devianto 2022).

Tahyudi develop a web-based application to predict flood events in Cilacap Regency using SVM and RF algorithms for classification. They find that the SVM model significantly outperformed RF, achieving an accuracy of 98.15%, making it the most reliable choice for flood prediction. The results indicate that the develop web-based system can enhance flood forecasting accuracy, providing a valuable tool for local disaster management efforts (Tahyudi 2022).

Surendran et al. develop a scene classification system using transfer learning with ResNet models to support disaster response and emergency management during flood-related incidents. They utilize a curated dataset of 200 flood-related scenes categorized into different emergency response scenarios and achieve classification accuracies of 88.88% with ResNet-50 and 91.04% with ResNet-101. Their findings suggest that the approach provides a promising tool for improving automated scene classification in disaster response (Surendran et al. 2022).

Although previous studies have explored deep learning-based approaches for flood segmentation, comparative analyses of UNet, SegNet, and DeepLabV3+ with different backbone architectures remain limited. Existing works have primarily focused on single-model evaluations or region-specific case studies, leaving a gap in understanding the generalization capabilities of different architectures across datasets. This study addresses this gap by conducting a systematic evaluation of UNet, SegNet, and DeepLabV3+ with EfficientNet and ResNet as feature extractors, providing valuable insights into their strengths and limitations for large-scale flood segmentation applications. Our findings contribute to enhancing AI-driven flood mapping methodologies, facilitating more effective disaster response and risk management strategies.

3 Materials and methods

This section describes the materials and methods used in this study, including dataset selection, preprocessing techniques, DL architectures, and evaluation metrics. The effectiveness of segmentation models largely depends on high-quality data preprocessing and appropriate architectural choices, both of which directly affect model accuracy and generalizability.

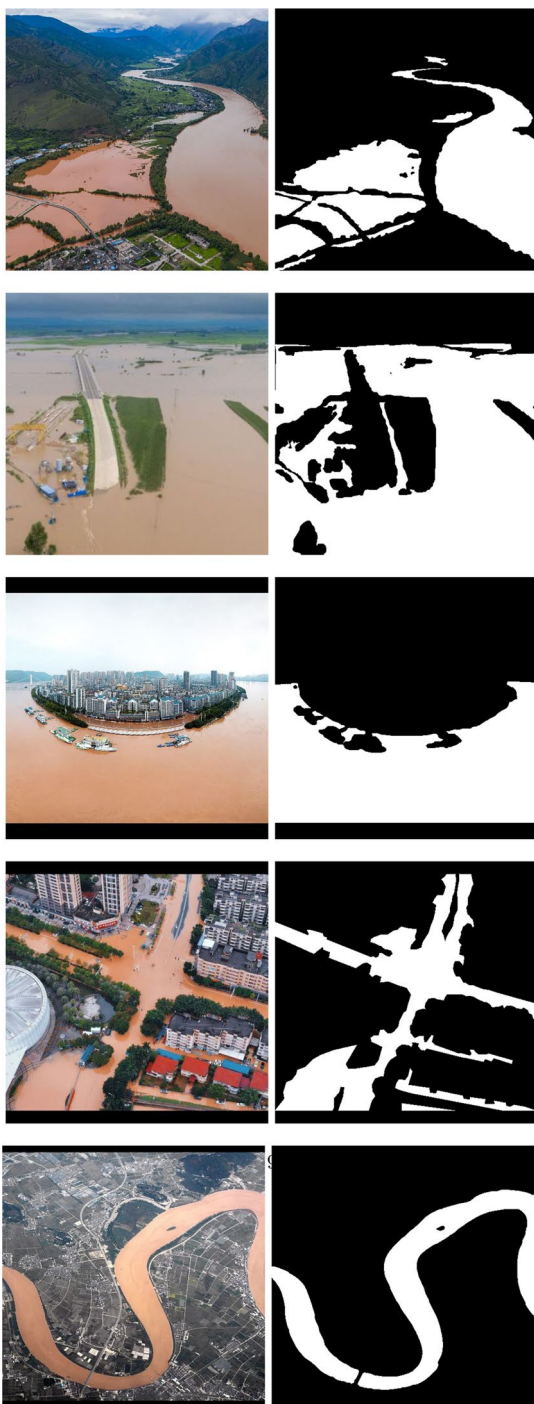
3.1 Materials

Labeled flood imagery datasets and high-performance computational resources employ to train and evaluate deep learning-based flood segmentation models. The dataset provides annotated images that enable the models to distinguish between flooded and non-flooded regions, forming the foundation for accurate segmentation.

3.1.1 Dataset

Kaggle provide the dataset utilized in this investigation ((Kaggle, 01.16.24/01.16.25) 01.16.24/01.16.25). The dataset consists of 663 images of flood-affected areas, each with a corresponding mask. The images have a resolution of 512×512 pixels and capture flooded regions from various angles and perspectives. Each image is paired with a segmentation mask, where flooded areas are labeled as positive pixels, while non-flooded areas remain unmarked. Figure 1 presents sample images from the dataset, showcasing both the original flood scenes and their corresponding labeled masks. This dataset plays a pivotal role in benchmarking deep learning-based flood segmentation models, enabling a structured evaluation of different architectures and feature extractors.

Fig. 1 Examples of original and masked images



3.1.2 Data preprocessing

The dataset comprises 663 flood-affected images, each paired with a corresponding segmentation mask. To ensure compatibility with the DL models and facilitate efficient training, a structured preprocessing pipeline is applied to both images and masks. Fig. 2 shows the overall data preparation workflow using the Flood Semantic Segmentation Dataset (FSSD).

A 5-fold cross-validation strategy was adopted to improve the robustness and reliability of the evaluation. In each fold, the dataset was divided into training and validation subsets, ensuring that the model was trained and evaluated on different partitions of the data. This approach provides a more comprehensive assessment of model generalizability.

Initially, all images and masks were resized to 256×256 pixels to ensure consistency in input dimensions across segmentation models. Pixel values were normalized to the $[0,1]$ range to standardize the data and improve numerical stability during training. The masks were binarized using a threshold of 0.5 to distinguish between flooded and non-flooded regions. An additional channel was added to the masks to make their shape $(256, 256, 1)$, as required by DL frameworks.

To further enhance generalization and avoid overfitting, data augmentation was applied using the Albumentations library. Augmentation techniques included horizontal flipping, rotation, translation, scaling, and brightness/contrast adjustments. The augmented dataset—comprising both original and transformed images—was used across all folds to ensure robust model training. Figure 2 illustrates the preprocessing pipeline.

3.2 Method

Flood segmentation is a complex task due to variations in flood extent, terrain conditions, and environmental factors. To address this, deep learning-based segmentation architectures are employed to accurately delineate flooded regions from satellite and aerial imagery.

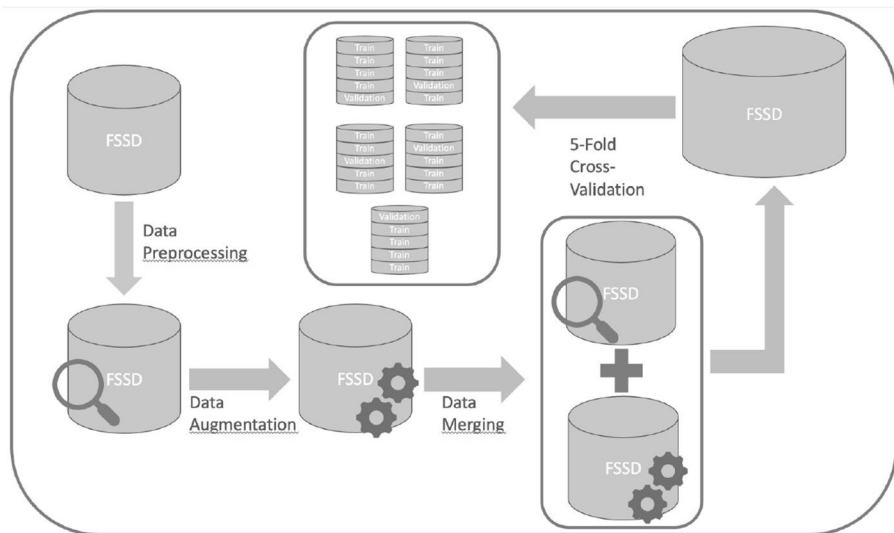


Fig. 2 Data preprocessing steps

These models effectively capture spatial and contextual information, improving segmentation performance compared to conventional techniques.

To achieve this, three widely used DL architectures –UNet, SegNet, and DeepLabV3+– are employed. These architectures have been widely adopted in semantic segmentation tasks due to their ability to capture both local and global contextual information. Furthermore, EfficientNet and ResNet are selected as backbone feature extractors to enhance feature representation and improve segmentation accuracy. EfficientNet is known for its optimized parameter efficiency and strong performance in image analysis tasks, while ResNet excels in capturing deep hierarchical features through residual learning. By integrating these architectures, the goal is to improve segmentation performance while maintaining computational efficiency.

3.2.1 Deep learning architectures for flood segmentation

Three widely used DL architectures –UNet, SegNet, and DeepLabV3+– are employed to perform flood segmentation. As these models can extract both local and global contextual characteristics, they have been widely used in semantic segmentation tasks.

A fully convolutional neural network called UNet was created specifically for the segmentation of biological images. UNet utilizes extensive data augmentation techniques to successfully train despite a limited amount of labeled data. The architecture has an expanding path for high-precision positioning and a shrinking path for gathering contextual data. This structure enables seamless segmentation of large images. Ronneberger et al. experimentally demonstrate that UNet performs high-accuracy segmentation on neuronal structures obtained with electron microscopy and cell images recorded using light microscopy. The main reason for UNet's superior performance in segmentation are its fully convolutional structure, balanced use of contracting and expanding paths, data augmentation techniques, customized loss function, and tiling strategy. While high-resolution segmentation maps are created by preserving location information thanks to the skip connection mechanism, the generalization ability of the model is increased with data augmentation. In addition, the special loss function that gives weight to boundary regions provides better separation of objects close to each other. The overlap-tile strategy and fast processing capacity of dividing large images make UNet an effective method in image segmentation (Ronneberger et al. 2015).

SegNet, a deep convolutional encoder-decoder architecture, has demonstrated its effectiveness in the field of image segmentation. SegNet uses a VGG16-based encoder network to extract low-resolution feature maps from the input image and then performs pixel-level classification with a decoder network that reuses the pooling indices to transform these feature maps back to the original resolution. This approach is computationally efficient due to the non-linear upsampling that does not require learning and offers lower memory requirements compared to other methods. The paper compares the performance of SegNet with common segmentation architectures such as Fully Convolutional Network (FCN), DeepLab-LargeFOV, and DeconvNet, and analyzes the trade-off between memory consumption, processing time, and accuracy. Particularly, experimental evaluations on CamVid road scenes and SUN RGB-D indoor scenes reveal that SegNet provides high class separation accuracy and is one of the most memory-efficient architectures. The main reasons for SegNet's superior performance in segmentation are its memory-efficient encoder-decoder architecture, learning-free upsampling method, and ability to preserve boundary details. While

the encoder network captures contextual information, the parser network upsamples using the indices saved during max-pooling so that the boundary details are better preserved and small objects are segmented more accurately. SegNet is appropriate for large datasets and applications with minimal memory requirements due to its advantages in terms of memory and computational efficiency. Experiments on CamVid and SUN RGB-D datasets prove that SegNet is a fast, accurate, and memory-friendly segmentation model (Badrinarayanan et al. 2017).

The DeepLabV3+ model is presented as an encoder-decoder architecture developed to increase the accuracy and efficiency in image segmentation. The difference between DeepLabV3+ and its previous version, DeepLabV3, is that DeepLabV3+ captures object boundaries better by adding a simple but effective decoder module. Atrous Spatial Pyramid Pooling (ASPP) module is improved to increase the ability to capture multi-scale context information, and depthwise separable convolutions are used in both ASPP and solver modules to make the model faster and more powerful. Experiments show that DeepLabV3+ achieves higher accuracy compared to previous methods on PASCAL VOC 2012 and Cityscapes datasets and exhibits high performance without post-processing. The main reason for the superior performance of DeepLabV3+ in segmentation are its encoder-decoder architecture, multi-scale context capture with Atrous Spatial Pyramid Pooling (ASPP), computational efficiency with depthwise separable convolutions, and the use of high-resolution feature maps. The decoder module increases the accuracy by segmenting object boundaries more sharply, while ASPP captures contextual information at different scales. Providing high accuracy without requiring post-processing and being a fast and lightweight model with its Xception-based structure makes DeepLabV3+ more efficient and successful compared to previous methods. Tests on the PASCAL VOC 2012 and Cityscapes datasets show that the model provides high accuracy in segmentation by better preserving object boundaries (Chen et al. 2018).

Flood segmentation requires a DL model capable of accurately delineating flooded and non-flooded areas while preserving spatial and boundary details. Due to their demonstrated efficacy in picture segmentation tasks—specifically, in collecting fine-grained details and managing intricate spatial variations—UNet, SegNet, and DeepLabV3+ were chosen for this work.

UNet was chosen for its fully convolutional structure, skip connections, and data augmentation adaptability, which enhance its ability to retain high-resolution details while generalizing well to diverse flood scenarios (Ronneberger et al. 2015). The skip connections allow the network to efficiently segment water bodies with irregular boundaries by combining low- and high-level spatial features, making it ideal for flood segmentation tasks.

SegNet was selected due to its memory-efficient encoder-decoder architecture and computational efficiency (Badrinarayanan et al. 2017). In large-scale flood segmentation, where high-resolution satellite and aerial images must be processed efficiently, SegNet provides an optimal trade-off between segmentation accuracy and computational cost. Its learning-free upsampling approach helps preserve boundary details, ensuring more precise segmentation of flooded areas.

DeepLabV3+ was incorporated for its multi-scale context capturing capability with Atrous Spatial Pyramid Pooling (ASPP) and high boundary delineation accuracy (Chen et al. 2018). Flooded regions vary in shape and scale, making ASPP crucial for capturing spatial information across different resolutions.

This study aims to develop a robust flood segmentation framework using the high-resolution segmentation of UNet, the computational efficiency of SegNet, and the multi-scale contextual awareness of DeepLabV3+. These models increase the ability to accurately identify flooded and non-flooded areas, providing precise segmentation for flood extent mapping and analysis.

3.2.2 Backbone networks for feature extraction

By removing hierarchical information from input images, backbone networks are essential to deep learning-based segmentation models. In this study, ResNet and EfficientNet were chosen as feature extractors to improve segmentation accuracy and computational efficiency. These backbone networks replace the standard encoders in UNet, SegNet, and DeepLabV3+, enhancing their ability to capture spatial and contextual information.

ResNet, or Residual Network, was introduced by He et al. in 2016. Addressing the challenges encountered in optimizing deep networks, it suggests that instead of learning a function directly, learning the residual function between the input and the target output provides a more efficient approach. In this direction, residual connections are added to the network architecture, allowing some layers to directly transmit the input data. Experiments show that this approach facilitates the optimization of deeper networks and achieves higher accuracy rates, especially on ImageNet and CIFAR-10 datasets. It has been demonstrated that the proposed 152-layer residual network (ResNet-152) has lower computational complexity than traditional deep networks such as VGG but exhibits superior performance. They also state that this method won first place in the 2015 COCO segmentation competition (He et al. 2016). It shows that ResNet can also be used as an effective feature extractor (backbone) for segmentation.

EfficientNet, introduced by Tan and Le in 2019, is a family of CNN designed for image classification and other computer vision tasks. Stating that existing methods usually only scale one of the singular dimensions, such as depth, width, or resolution, the study proposes a new methodology called compound scaling. By employing fixed multipliers to scale all network dimensions in a balanced manner, this technique seeks to enhance optimization, lower computing costs, and improve model accuracy. Experimental results show that EfficientNet provides higher accuracy than modern CNN architectures such as ResNet, Inception, and NasNet and is also significantly more efficient with an 8.4 times smaller model size and a 6.1 times faster inference time. It is also shown that EfficientNet is successful in transfer learning processes and achieves best-of-the-art accuracy rates on various datasets (Tan and Le 2019). It indicates that EfficientNet can also be used as an effective feature extractor (backbone) in other visual perception tasks such as segmentation.

By incorporating these backbone networks, this study aims to leverage ResNet's deep feature extraction capability and EfficientNet's optimized architecture to develop a high-performance flood segmentation framework. The combination of these architectures ensures that the segmentation models maintain high accuracy while efficiently processing large-scale remote sensing data.

3.2.3 Training configuration and implementation

Prior to training, all input images and corresponding segmentation masks were resized to 256×256 pixels and normalized to the $[0, 1]$ range. The segmentation masks were binarized using a threshold value of 0.5. To enhance model generalization and prevent overfitting, data augmentation was applied using the Albumentations library. Augmentation strategies included random horizontal flipping, rotation, translation, scaling, and brightness/contrast adjustment.

A 4-fold cross-validation strategy was adopted to improve the robustness and reliability of the evaluation. Each fold was trained and validated independently, and the final performance metrics were averaged over all folds.

U-Net with EfficientNet-B0 backbone is implemented as an encoder-decoder architecture, where the encoder is initialized with the convolutional layers of EfficientNet-B0 pre-trained on ImageNet. The decoder path consists of five upsampling blocks, each followed by convolution and ReLU activation. Skip connections are used to merge encoder features with decoder outputs at multiple levels, preserving spatial detail. The model outputs a single-channel sigmoid-activated segmentation mask. This configuration enables efficient feature extraction with reduced parameter count while maintaining high-resolution predictions.

U-Net with ResNet50 backbone utilizes the pretrained ResNet50 network as an encoder to extract hierarchical features. The architecture includes a symmetric decoder path composed of four upsampling blocks, each followed by skip connections that merge features from corresponding encoder layers. After the final decoding stage, a convolutional block generates a single-channel sigmoid-activated output mask. The model benefits from ResNet's deep feature extraction capacity and U-Net's spatial precision, enabling accurate delineation of flood regions.

The proposed DeepLabV3+ architecture with EfficientNet-B0 backbone employs a pre-trained EfficientNet-B0 model as a feature extractor, leveraging its lightweight and high-performing representation capacity. The encoder output is processed by an Atrous Spatial Pyramid Pooling (ASPP) module that captures multi-scale contextual information using dilated convolutions and global average pooling. This is followed by a custom decoder consisting of multiple upsampling, convolutional, batch normalization, and dropout layers to reconstruct the segmentation mask. The final output is generated through a sigmoid-activated convolution layer. This design effectively balances feature richness and computational efficiency, making it suitable for near-real-time flood segmentation.

We implemented a DeepLabV3+ segmentation architecture using a ResNet50 backbone, an Atrous Spatial Pyramid Pooling (ASPP) module, and a multi-stage decoder to capture both contextual and spatial information. The model extracts high-level features using the ResNet50 encoder and processes them through the ASPP block to increase the receptive field with multi-scale context. The decoder reconstructs the segmentation map through progressive upsampling and convolutional layers, followed by a sigmoid activation for binary mask prediction.

We developed a custom segmentation model by integrating the SegNet architecture with an EfficientNet-B0 encoder. The encoder is constructed using pretrained EfficientNet blocks to extract hierarchical features at multiple levels, while the decoder mirrors the encoder with transposed convolutional layers for upsampling. Unlike U-Net, this design does not include

skip connections between encoder and decoder stages. The final segmentation mask is produced through a sigmoid-activated output layer for binary classification.

We developed a segmentation model based on the SegNet architecture, utilizing ResNet50 as the encoder. The pretrained ResNet50 backbone was modified by removing the average pooling and fully connected layers, and the feature maps from the final convolutional block were used in a symmetric decoder composed of upsampling and convolutional layers. This architecture does not include skip connections, remaining faithful to the classical SegNet design.

All segmentation models in this study were implemented using the PyTorch deep learning framework. To ensure a fair and consistent comparison across architectures and backbone networks, all models were trained using the same hyperparameter settings. Specifically, each model was trained for 25 epochs with a batch size of 8. The Adam optimizer was employed with a fixed learning rate of $1e-4$, and Binary Cross Entropy Loss (BCELoss) was used for the binary segmentation task.

To support transparency and reproducibility, all training scripts, model definitions, and configuration files are publicly available via GitHub link at the Source Code section.

3.3 Model evaluation

Five widely used segmentation metrics—accuracy, precision, recall, F1-score and IoU—are utilized to thoroughly assess the pre-trained models' performance. These metrics assess how effectively the model distinguishes flooded and non-flooded regions in remote sensing imagery. The mathematical formulations for these metrics are as follows:

3.3.1 Accuracy

The most crucial parameter for evaluating classification models is accuracy, which informs us how well the algorithm learns the dataset's data and how well it predicts the test data (Agarwal et al. 2021). It is shown in Eq. (1).

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

True Positive (TP): The quantity of accurately anticipated flooded pixels. *True Negative (TN)*: The quantity of non-flooded pixels that were accurately anticipated. *False Positive (FP)*: The quantity of pixels that are not flooded but are mistakenly labeled as flooded (false alarms). *False Negative (FN)*: Inaccurately identifying flooded pixels as non-flooded (missed detections).

3.3.2 Precision

The precision metric provides insight into how well the model can detect good predictions while lowering the possibility of false positives (Bahrami and Arbabkhah 2024). It is shown in Eq. (2).

$$Precision = \frac{TP}{TP + FP} \quad (2)$$

3.3.3 Recall

The recall measure shows how well the model can recognize every good example of a given class (Zhang et al. 2024). It is shown in Eq. (3).

$$Recall = \frac{TP}{TP + FN} \quad (3)$$

3.3.4 F1-score

F1-score is a metric used to estimate model performance. To find the balance between Precision and Recall, the harmonic mean of these two values is taken (Saheed and Arowolo 2021; Usha et al. 2021) and expressed by Eq. (4)

$$F1 - Score = 2 \frac{Precision \times Recall}{Precision + Recall}. \quad (4)$$

3.3.5 Intersection over union(IoU)

The intersection-over-union (IoU) is usually used to measure the performance of any object category segmentation method (Rahman and Wang 2016). It is shown in Eq. (5).

$$IoU = \frac{TP}{TP + FN + FP} \quad (5)$$

In this study, the results of UNet, SegNet, and DeepLabV3+ models were compared using EfficientNet and ResNet methods as backbones.

4 Experimental results and discussion

An analysis of Table 1 and Table 2 reveals that UNet with an EfficientNet backbone achieves the highest segmentation performance among all tested models, with an accuracy of 0.9705, a precision of 0.9609, a recall of 0.9548, an F1-score of 0.9575 and an IoU of 0.9271. This suggests that EfficientNet enhances UNet's ability to accurately classify pixels while maintaining an optimal balance between precision and recall. The superior performance of UNet with EfficientNet indicates its effectiveness in capturing fine-grained spatial features, making it the most reliable choice for flood segmentation in this study.

Although UNet with ResNet achieves competitive performance, with an accuracy of 0.9675, a precision of 0.9597, a recall of 0.9511, an F1-score of 0.9554 and an IoU of

Table 1 Comparison of models for segmentation: backbone as efficientNet model

Model	Accuracy	Precision	Recall	F1-score	IoU
UNet	0.9705	0.9609	0.9548	0.9575	0.9271
SegNet	0.9414	0.9223	0.9173	0.9195	0.8459
DeepLabV3+	0.9344	0.9063	0.9148	0.9105	0.8358

Table 2 Comparison of models for segmentation: backbone as ResNet model

Model	Accuracy	Precision	Recall	F1-score	IoU
UNet	0.9675	0.9597	0.9511	0.9554	0.9113
SegNet	0.9614	0.9432	0.9455	0.9439	0.8950
DeepLabV3+	0.9147	0.8848	0.8814	0.8830	0.7906

**Fig. 3** Results obtained from using the efficientNet backbone in the UNet model (Input, Original Mask, Predicted Mask)

0.9113, it falls slightly short of UNet with EfficientNet. This suggests that while ResNet provides strong feature extraction, it does not surpass EfficientNet's optimization in segmentation tasks.

SegNet with EfficientNet performs reasonably well, with an accuracy of 0.9414, a precision of 0.9223, a recall of 0.9173, an F1-score of 0.9195 and an IoU of 0.8459. However, its accuracy remains lower than that of UNet, suggesting that while it can effectively segment flooded regions, it lacks the finer detail retention capability of UNet. SegNet with ResNet, on the other hand, demonstrates higher precision (0.9614), recall (0.9432), accuracy (0.9455), F1-score (0.9439) and IoU (0.8950) compared to its EfficientNet counterpart. DeepLabV3+ with EfficientNet achieves moderate performance, with an accuracy of 0.9344, a precision of 0.9063, a recall of 0.9148, an F1-score of 0.9105 and an IoU of 0.8358. The slightly lower recall compared to SegNet with EfficientNet suggests that DeepLabV3+ might be missing some true positive instances, leading to under-segmentation in certain cases. Furthermore, DeepLabV3+ with ResNet exhibits the lowest accuracy (0.9147), precision (0.8848), recall (0.8814), F1-score (0.8830) and IoU (0.7906) among the models in the ResNet group, making it less effective in terms of both correctly identifying flood-affected pixels and minimizing false negatives.

These results highlight that UNet with EfficientNet consistently outperforms other models in terms of segmentation accuracy, indicating that the combination of UNet's architecture and EfficientNet's optimized feature extraction provides the most effective approach for flood segmentation. In contrast, SegNet with ResNet provides a good mix between accuracy and precision, which makes it a good substitute in situations when recall is not as important as high precision (Figs. 3 and 4).

Table 3 presents a comparative analysis of segmentation performance between our study and the results reported in Simantiris and Panagiotakis's study (Simantiris and Panagiotakis 2024). Both studies utilize the same flood image dataset, ensuring a fair basis for comparison. Accuracy, precision, and recall are among the important evaluation parameters that form the basis of the comparison. Despite using the same dataset, the results obtained in

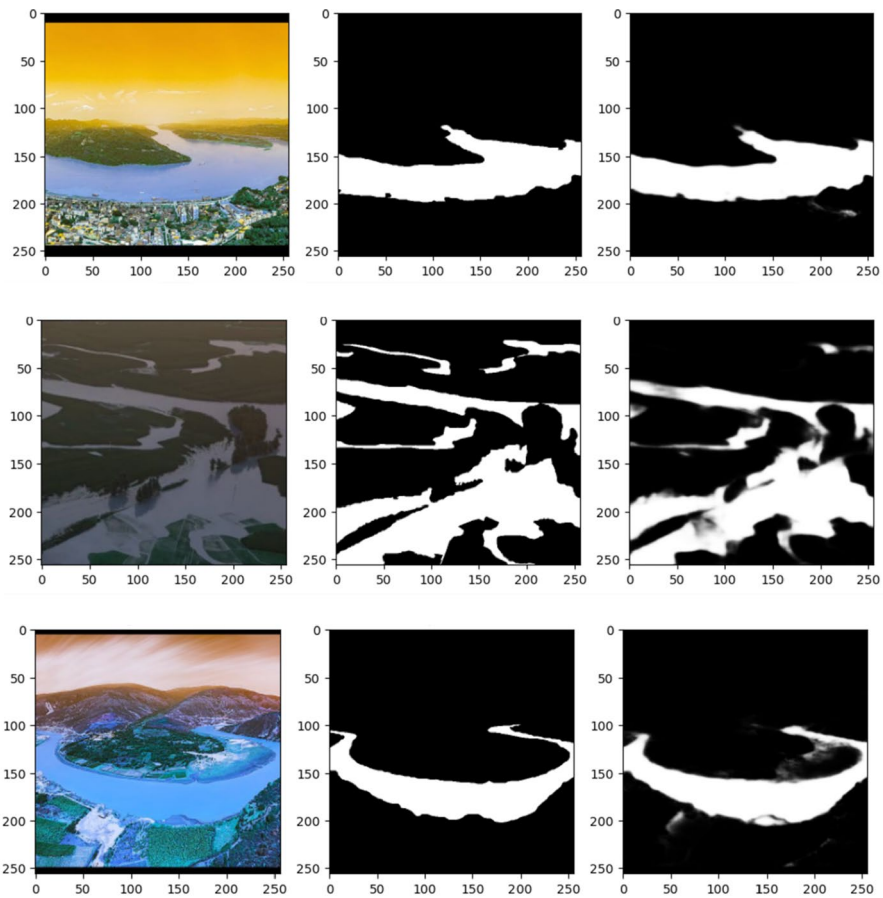


Fig. 4 Results obtained from using the ResNet backbone in the UNet model (Input, Original Mask, Predicted Mask)

Table 3 Comparison of segmentation performance between our study and Simantiris and Panagiotakis's study (Simantiris and Panagiotakis 2024)

Model	Backbone	Accuracy	Precision	Recall	F1-Score	IoU
UNet	EfficientNet	0.9705	0.9609	0.9548	0.9575	0.9271
UNet	ResNet	0.9675	0.9597	0.9511	0.9554	0.9113
SegNet	EfficientNet	0.9414	0.9223	0.9173	0.9195	0.8459
SegNet	ResNet	0.9614	0.9432	0.9455	0.9439	0.8950
DeepLabV3+	EfficientNet	0.9344	0.9063	0.9148	0.9105	0.8358
DeepLabV3+	ResNet	0.9147	0.8848	0.8814	0.8830	0.7906
UFS-HT-REM	—	0.849	0.795	0.786	0.791	—

(Simantiris and Panagiotakis 2024)

our study demonstrate notably higher accuracy and precision. Differences in preprocessing approaches, such as data augmentation, picture normalization, and scaling techniques, are responsible for this improvement. These strategies may have assisted in improving feature representation and model generalization. These enhancements boost segmentation accuracy by enhancing the model's ability to distinguish between flooded and non-flooded regions. The findings indicate that UNet with EfficientNet backbone achieves the highest overall accuracy (0.9705) and precision (0.9609), significantly outperforming the UNet model reported by Simantiris and Panagiotakis's study (Simantiris and Panagiotakis 2024), which achieved an accuracy of 0.8974 and precision of 0.8281. This indicates that EfficientNet is a powerful backbone for feature extraction, improving segmentation performance by effectively capturing detailed spatial characteristics in flooded areas.

Similarly, DeepLabV3+ with EfficientNet backbone achieves an accuracy of 0.9344, surpassing the UFS-HT-REM model from Simantiris and Panagiotakis's study (Simantiris and Panagiotakis 2024), which achieved 0.849 accuracy. The improved performance can be attributed to the integration of EfficientNet as a feature extractor, which enhances deep feature learning while maintaining computational efficiency. Furthermore, SegNet models exhibit strong precision and accuracy, particularly when combined with ResNet as a backbone, indicating that ResNet enhances pixel-wise classification accuracy by extracting more detailed contextual features. These results demonstrate that our approach, which integrates EfficientNet and ResNet into UNet, SegNet, and DeepLabV3+, significantly improves segmentation accuracy and precision in flood mapping applications. The variations in performance metrics between our study and Simantiris and Panagiotakis's study (Simantiris and Panagiotakis 2024) can be attributed to differences in: Backbone networks: Our study employs EfficientNet and ResNet, whereas Simantiris and Panagiotakis's study (Simantiris and Panagiotakis 2024) primarily use standard CNN-based feature extractors. Preprocessing techniques: Differences in image normalization, data augmentation strategies, and resizing methods may have influenced segmentation outcomes. Model architectures and hyperparameters: The optimization techniques and training settings (e.g., learning rate, batch size, number of epochs) likely contributed to performance variations. In conclusion, the integration of EfficientNet and ResNet as backbone networks demonstrates superior segmentation performance in flood mapping tasks, highlighting the effectiveness of deep feature extraction in distinguishing flood-affected regions.

5 Conclusions

This study demonstrates how effectively deep learning-based flood segmentation algorithms define areas affected by flooding. The dataset used includes 663 images taken from the same perspective, featuring largely uniform scenes. This homogeneity may limit the generalizability of the results to diverse flood scenarios. Among the evaluated architectures, UNet with an EfficientNet backbone achieves the highest accuracy of 0.9705, precision of 0.9609, recall of 0.9548, F1-score of 0.9575 and IoU of 0.9271, making it the most successful model for flood segmentation. According to findings, EfficientNet performs better than other feature extractors for segmentation tasks, especially where overall accuracy and striking a balance between precision and recall are important considerations. Beyond model performance, this research shows the practical applicability of deep learning-based

flood segmentation for disaster response and management. Unlike traditional hydrological and remote sensing-based flood mapping approaches, which often require manual intervention and are computationally expensive, the proposed method offers a fully automated and scalable solution. By leveraging EfficientNet's lightweight yet powerful architecture, the segmentation models can efficiently process large-scale remote sensing imagery, making them suitable for real-time flood monitoring and risk assessment. This capability is particularly crucial for emergency response teams, enabling faster decision-making and allocating resources during floods.

Compared to previous studies, our approach improves segmentation accuracy by integrating advanced backbone networks, enhancing both feature extraction and computational efficiency. The use of EfficientNet and ResNet backbones allows the model to record intricate spatial details while maintaining a low computational footprint, making it a feasible choice for deployment in cloud-based or edge-computing environments.

Future research may focus on optimizing deep learning models for flood detection, addressing data scarcity issues, and developing computationally efficient architectures that can be deployed in resource-limited environments. Besides, the generalization capability of the model could be improved by incorporating datasets comprising images from different flood regions (with the same technical characteristics), or by combining datasets with varying properties in terms of resolution or image size. By combining these technological innovations with existing hydrological and hydraulic models and different datasets, flood management strategies can be significantly improved, leading to faster response times, reduced economic losses, and more effective disaster mitigation efforts (Mou et al. 2025).

Funding Open access funding provided by the Scientific and Technological Research Council of Türkiye (TÜBİTAK). This study was completed without any funding, grants, or support from public, commercial, or non-profit organizations.

Data availability <https://www.kaggle.com/datasets/lihuayang111265/flood-semantic-segmentation-dataset>

Code availability <https://github.com/seydakarc/Flood-segmentation/tree/main>

Declarations

Conflict of interest The authors declare that there is no Conflict of interest regarding the publication of this paper.

Ethical approval There is no ethical conflict.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by/4.0/>.

References

- Agarwal A, Sharma P, Alshehri M et al (2021) Classification model for accuracy and intrusion detection using machine learning approach. *PeerJ Comput Sci* 7:e437
- Apene OZ, Hampo JA, Ogeh CO, Hussein SU (2023) Flood endangered area classification using the k-nearest neighbour algorithm. *Sciences* 1(5):1051–1061. <https://doi.org/10.59324/ejtas.2023.1%285%29.92>
- Badrinarayanan V, Kendall A, Cipolla R (2017) Segnet: a deep convolutional encoder-decoder architecture for image segmentation. *IEEE Trans Pattern Anal Mach Intell* 39(12):2481–2495
- Bahrami B, Arbabkhah H (2024) Enhanced flood detection through precise water segmentation using advanced deep learning models. *J Civ Eng Res* 6(1):1–8
- Bauer-Marschallinger B, Cao S, Tupas ME et al (2022) Satellite-based flood mapping through bayesian inference from a sentinel-1 sar datacube. *Remote Sens* 14(15):3673
- Bentivoglio R, Isufi E, Jonkman SN et al (2022) Deep learning methods for flood mapping: a review of existing applications and future research directions. *Hydrol Earth Syst Sci Discuss* 2022:1–50
- Caleb A, Adeola FR. AI-driven flood management systems. 2024. https://www.researchgate.net/publication/387516698_AIDriven_Flood_Management_Systems
- Chamatidis I, Istrati D, Lagaros ND (2024) Vision transformer for flood detection using satellite images from sentinel-1 and sentinel-2. *Water* 16(12):1670
- Chebii SJ, Mukolwe MM, Ong'or BI (2022) River flow modelling for flood prediction using artificial neural network in ungauged perkerra catchment, baringo county, kenya. *Water Pract Technol* 17(4):914–929
- Chen LC, Zhu Y, Papandreou G, et al (2018) Encoder-decoder with atrous separable convolution for semantic image segmentation. In: *Proceedings of the European conference on computer vision (ECCV)*, pp 801–818
- Du W, FitzGerald GJ, Clark M et al (2010) Health impacts of floods. *Prehosp Disaster Med* 25(3):265–272
- Dwiasnati S, Devianto Y (2022) Optimization of flood prediction using svm algorithm to determine flood prone areas. *J Syst Eng Inf Technol (JOSEIT)* 1(2):40–46 (<https://api.semanticscholar.org/CorpusID:257140792>)
- Elstohy R, Ali EM (2023) A flash flood detected area using classification-based image processing for sentinel-2 satellites data: a case study of zafaraana road at red sea. *The Egypt J Remote Sens Space Sci* 26(3):807–814 (<https://api.semanticscholar.org/CorpusID:261156852>)
- Foroughnia F, Alfieri SM, Menenti M et al (2022) Evaluation of sar and optical data for flood delineation using supervised and unsupervised classification. *Remote Sens* 14(15):3718 (<https://api.semanticscholar.org/CorpusID:251358177>)
- Ganjirad M, Delavar M (2023) Flood risk mapping using random forest and support vector machine. *ISPRS Annals of the Photogramm Remote Sens Spat Inf Sci* 10:201–208
- Ghosh B, Garg S, Motagh M et al (2024) Automatic flood detection from sentinel-1 data using a nested unet model and a nasa benchmark dataset. *PFG-J Photogramm Remote Sens Geoinf Sci* 92(1):1–18
- He K, Zhang X, Ren S, et al (2016) Deep residual learning for image recognition. In: *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp 770–778
- Hemmati M, Ellingwood BR, Mahmoud HN (2020) The role of urban growth in resilience of communities under flood risk. *Earth's Future* 8(3):e2019EF001382
- Hernández D, Cecilia JM, Cano JC et al (2022) Flood detection using real-time image segmentation from unmanned aerial vehicles on edge-computing platform. *Remote Sens* 14(1):223
- Kaggle LY (01.16.24/01.16.25) Flood semantic segmentation dataset. <https://www.kaggle.com/datasets/lihuayang111265/flood-semantic-segmentation-dataset>, accessed: 01.16.25
- Kalidhas M, Sivakumar R (2022) Image processing and supervised classification of landsat data for flood impact assessment on land use and land cover. In: *2022 2nd International Conference on Technological Advancements in Computational Sciences (ICTACS)* pp 437–440. <https://api.semanticscholar.org/CorpusID:255187936>
- Kim D, Park JP, Han H, et al (2023) Application of ai-based models for flood water level forecasting and flood risk classification. *KSCE J Civ Eng* 27:3163–3174. <https://api.semanticscholar.org/CorpusID:258537818>
- Kumar V, Sharma KV, Caloiero T et al (2023) Comprehensive overview of flood modeling approaches: a review of recent advances. *Hydrology* 10(7):141
- Melgar-García L, Martínez-Álvarez F, Bui DT et al (2023) A novel semantic segmentation approach based on u-net, wu-net, and u-net++ deep learning for predicting areas sensitive to pluvial flood at tropical area. *Int J Digital Earth* 16:3661–3679 (<https://api.semanticscholar.org/CorpusID:261845320>)
- Mou SA, Chowdhury TN, Mannan AI, et al (2025) AI driven water segmentation with deep learning models for enhanced flood monitoring. *arXiv preprint arXiv:2501.08266*

- Munawar HS, Hammad AW, Waller ST (2021) A review on flood management technologies related to image processing and machine learning. *Autom Constr* 132:103916
- Nouwakpo S, Bjorneberg D, McGwire K et al (2024) Mapping irrigation methods in the northwestern us using deep learning classification. *Water Resour Res* 60(8):e2023WR036155
- Patil V, Khadke Y, Joshi A et al (2024) Flood mapping and damage assessment using ensemble model approach. *Sens Imaging* 25(1):15
- Pech-May F, Sanchez-Hernández JV, López-Gómez LA et al (2023) Flooded areas detection through sar images and u-net deep learning model. *Computación y Sistemas* 27(2):449–458
- Puttinaovarat S, Saeliw A, Pruitikanee S et al (2022) Flood damage assessment geospatial application using geoinformatics and deep learning classification. *Int J Interact Mob Technol* 16:71–97 (<https://api.semanticscholar.org/CorpusID:253559084>)
- Rahman MA, Wang Y (2016) Optimizing intersection-over-union in deep neural networks for image segmentation. In: *International symposium on visual computing*, Springer, pp 234–244
- Ronneberger O, Fischer P, Brox T (2015) U-net: Convolutional networks for biomedical image segmentation. In: *Medical image computing and computer-assisted intervention—MICCAI 2015: 18th international conference, Munich, Germany, October 5–9, 2015, proceedings, part III* 18, Springer, pp 234–241
- Saheed YK, Arowolo MO (2021) Efficient cyber attack detection on the internet of medical things-smart environment based on deep recurrent neural network and machine learning algorithms. *IEEE Access* 9:161546–161554
- Sellami EM, Rhinane H (2024) Google earth engine and machine learning for flash flood exposure mapping-case study: Tetouan, morocco. *Geosciences* 14(6):152
- Simantiris G, Panagiotakis C (2024) Unsupervised color-based flood segmentation in uav imagery. *Remote Sens* 16(12):2126
- Sudiana D, Riyanto I, Rizkinia M, et al (2023) Development of three-dimensional convolutional neural network for urban flood classification using synthetic aperture radar multi-temporal image. In: *2023 8th Asia-Pacific Conference on Synthetic Aperture Radar (APSAR)* pp 1–6. <https://api.semanticscholar.org/CorpusID:267044187>
- Surendran R, Anitha J, Angelopoulou A, et al (2022) Transfer learning based natural scene classification for scene understanding by intelligent machines. In: *International Conference on Conceptual Structures*, <https://api.semanticscholar.org/CorpusID:250076700>
- Tahyudi I (2022) Website-based application for flood event prediction using machine learning method in cilacap district. *Telematika* 15(1):48–56 (<https://api.semanticscholar.org/CorpusID:247201001>)
- Tan M, Le Q (2019) Efficientnet: rethinking model scaling for convolutional neural networks. In: *International conference on machine learning*, PMLR, pp 6105–6114
- Usha G, Narang M, Kumar A (2021) Detection and classification of distributed dos attacks using machine learning. In: *Computer Networks and Inventive Communication Technologies: Proceedings of Third ICCNCT 2020*. Springer, p 985–1000
- Yolina H, Rojali Irwansyah E (2023) Development of flood-prone area classification program using linear classifier method based on geomorphic flood index and land cover. *Procedia Comput Sci* 216:396–405 (<https://api.semanticscholar.org/CorpusID:255661088>)
- Zhang S, Li W, Wang R et al (2024) Daliws: a high-resolution dataset with precise annotations for water segmentation in synthetic aperture radar images. *Remote Sens* 16(4):720